

①

a

Household:

$$\max_{C_1, l_1, C_2, l_2} \frac{C_1^{1-\sigma}}{1-\sigma} + \psi \ln(l_1) + \beta \left[\frac{C_2^{1-\sigma}}{1-\sigma} + \psi \ln(l_2) \right]$$

$$s.t.: L_t + l_t = 1$$

$$\text{Join} \begin{cases} C_1 + a \leq w_1 L_1 + r_1 \bar{K} + \Pi_1 \\ C_2 \leq a(1+R) + w_2 L_2 + r_2 \bar{K} + \Pi_2 \end{cases}$$

$$\boxed{C_1 + \frac{C_2}{1+R} \leq w_1 L_1 + r_1 \bar{K} + \Pi_1 + \frac{w_2 L_2 + r_2 \bar{K} + \Pi_2}{1+R}}$$

Inter-temporal BC

Firms (Same Problem in $t=1$ and $t=2$ - Static)

$$\max_{L_t, k_t} \Pi_t = A_t k_t^\alpha L_t^{1-\alpha} - w_t L_t - r_t k_t$$

$$\left. \begin{array}{l} w = \text{MPL} \\ r = \text{MPK} \end{array} \right\} \text{CRS + Perfect competition} \rightarrow \boxed{\Pi_t = 0}$$

⑤ Household:

$$L_t = 1 - l_t$$

$$\left. \begin{array}{l} C_1: C_1^{-\sigma} = \lambda \\ C_2: \beta \cdot C_2^{-\sigma} = \lambda \cdot \frac{1}{1+R} \end{array} \right\} \boxed{C_1^{-\sigma} = \beta(1+R) \cdot C_2^{-\sigma}} \quad \begin{array}{l} \text{Euler} \\ \text{equation} \end{array} \quad (1)$$

$$\left. \begin{array}{l} l_1: \frac{\psi}{l_1} = \lambda \cdot w_1 \\ l_2: \beta \cdot \frac{\psi}{l_2} = \lambda \cdot \frac{w_2}{1+R} \end{array} \right\} \begin{array}{l} w_1 \cdot \beta(1+R) \cdot \frac{1}{w_2} \cdot \frac{\psi}{l_2} = \frac{\psi}{l_1} \\ \boxed{\frac{w_1}{w_2} \cdot \beta(1+R) = \frac{l_2}{l_1}} \end{array} \quad \begin{array}{l} \text{Labor-leisure} \\ \text{inter temporal} \\ \text{condition} \end{array} \quad (2)$$

$$C_1^{-\sigma} = \frac{\psi}{l_1} \cdot \frac{1}{w_1} \rightarrow \boxed{\frac{C_1^{\sigma}}{l_1} = \frac{w_1}{\psi}} \quad \begin{array}{l} \text{Intra temporal} \\ \text{consumption-leisure tradeoff} \end{array} \quad (3)$$

Similar holds for $t=2 \rightarrow \boxed{\frac{C_2^{\sigma}}{l_2} = \frac{w_2}{\psi}} \quad (4)$

↳ Less relevant

$$\text{Or: } \frac{\psi \cdot C_t^{\sigma}}{l_t} = w_t$$

↳ MRS

Firms

$$w_t = (1-\alpha)A_t \left(\frac{k_t}{l_t}\right)^\alpha \text{ MPL}$$

$$r_t = \alpha \cdot A_t \left(\frac{k_t}{l_t}\right)^{\alpha-1} \text{ MPK}$$

© Allocation $\{c_t, l_t, L_t, k_t\}_{t=1}^\infty$ such that:

→ Households choices adhere to (1)-(4)

→ Firms choices adhere to MPL and MPK

→ Markets clear:

$$c_t = F(\bar{k}, L_t) \quad k_t = \bar{k} \quad L_t = 1 - l_t$$

Can also define in terms of max Problems to be solved

↳ Market-clearing conditions don't change

(d) $C_t = F(K_t, L_t)$ $w_t = (1-\alpha)A_t K_t^\alpha L_t^{1-\alpha}$

From (b): $\frac{C_t^\sigma}{l_t} = \frac{w_t}{\psi}$ $MRS = MRT \rightarrow \psi \frac{C_t^\sigma}{l_t} = w_t$

$L_t = 1 - l_t$

$$\psi \frac{[A_t K_t^\alpha L_t^{1-\alpha}]^\sigma}{1 - L_t} = (1-\alpha)A_t K_t^\alpha L_t^{1-\alpha}$$

Since $K_t = \bar{K}$
This eq. defines
 L_t (and l_t)

$\hookrightarrow L_t^*$

$$C_t = F(\bar{K}, L_t^*) \rightarrow C_1^\sigma = \beta(1+R)C_2^\sigma \rightarrow \left(\frac{C_2}{C_1}\right)^\sigma \cdot \frac{1}{\beta} = 1+R$$

$$\left(\frac{A_2 \bar{K}^\alpha L_2^{1-\alpha}}{A_1 \bar{K}^\alpha L_1^{1-\alpha}}\right)^\sigma \cdot \frac{1}{\beta} = 1+R \rightarrow \therefore$$

$$1+R = \frac{1}{\beta} \cdot \left[\frac{A_2}{A_1} \left(\frac{L_2}{L_1}\right)^{1-\alpha}\right]^\sigma$$

If $K \neq \bar{K}$ then
capital would also
matter, but problem
would change

©

$$\Psi \cdot \frac{[A_t K_t^\alpha L_t^{1-\alpha}]^\sigma}{1-L_t} = (1-\alpha) A_t K_t^\alpha L_t^{-\alpha}$$

$$\ln \Psi + \sigma \ln (A_t K_t^\alpha L_t^{1-\alpha}) - \ln (1-L_t) = \ln (1-\alpha) + \ln (A_t K_t^\alpha L_t^{-\alpha})$$

$$\sigma(1-\alpha) \ln L_t + \sigma \ln A_t - \ln (1-L_t) + \phi_0 = \ln A_t - \alpha \ln L_t + \phi_1$$

$$[\sigma(1-\alpha) + \alpha] \ln L_t - \ln (1-L_t) = (1-\sigma) \ln A_t + \phi_1 - \phi_0$$

$$\downarrow$$

$$d \ln (1-L_t) = \frac{L_t}{1-L_t} d \ln L_t$$

↳ irrelevant parts for diff

$$\therefore \frac{d \ln L_t}{d \ln A_t} = \frac{1-\sigma}{\sigma(1-\alpha) + \alpha + \frac{L_t}{1-L_t}}$$

Elasticity of labor (in eq) w.r.t. productivity

↳ Show it's not only firms' labor demand!

(2)

(a) $\max_{\{C_t, a_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma}}{1-\sigma}$

s.t:

$$C_t + a_{t+1} = w_t + (1+r_t) \cdot a_t$$

$\beta^t C_t^{-\sigma} = \lambda_t$ $\uparrow$ MU of C_t

$\beta^t \lambda_t = \beta^{t+1} (1+r_{t+1}) \lambda_{t+1}$ $\rightarrow$ MU of waiting until $t+1$

$\left. \begin{array}{l} \beta^t C_t^{-\sigma} = \lambda_t \\ \beta^t \lambda_t = \beta^{t+1} (1+r_{t+1}) \lambda_{t+1} \end{array} \right\} C_t^{-\sigma} = \beta \cdot (1+r_{t+1}) \cdot C_{t+1}^{-\sigma}$

$\frac{C_{t+1}}{C_t} = \left(\beta (1+r_{t+1}) \right)^{1/\sigma}$ OU

$\left(\frac{C_{t+1}}{C_t} \right)^{\sigma} = \beta (1+r_{t+1})$ Euler equation

$\sigma \rightarrow$ Intertemporal elasticity of substitution

$$\frac{C_{t+1}}{C_t} = \left(\beta (1+r_{t+1}) \right)^{1/\sigma} \rightarrow \ln(C_{t+1}/C_t) = \boxed{\frac{1}{\sigma}} \ln MRS$$

Consumption grows/changes due to relative differences in discount factor and interest-rate

$$\textcircled{b} \quad \Pi_t = A \cdot k_t^\alpha L_t^{1-\alpha} - w_t \cdot L_t - r_t^k k_t$$

$$\therefore w_t = F_L(k_t, L_t) = \alpha \cdot A \left(\frac{k_t}{L_t} \right)^{\alpha-1}$$

$$r_t^k = F_K(k_t, L_t) = (1-\alpha) A \cdot \left(\frac{k_t}{L_t} \right)^\alpha$$

$\textcircled{c}$

We're assuming $a_t \rightarrow k_t$, so $r_t = r_t^k$

Households:

$$BC: C_t + a_{t+1} + k_{t+1} = w_t + (1+r_t)a_t + (1+r_t^k - \delta)k_t$$

Solving again:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} + \lambda_t \cdot [w_t + (1+r_t)a_t + (1+r_t^k - \delta)k_t - C_t - a_{t+1} - k_{t+1}] \right]$$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}} \rightarrow \lambda_t (1+r_{t+1}) = \beta \cdot \lambda_{t+1}$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} \rightarrow \lambda_t (1+r_{t+1}^k - \delta) = \beta \cdot \lambda_{t+1}$$

$$\left. \begin{array}{l} \lambda_t (1+r_{t+1}) = \beta \cdot \lambda_{t+1} \\ \lambda_t (1+r_{t+1}^k - \delta) = \beta \cdot \lambda_{t+1} \end{array} \right\} 1+r_{t+1} = 1+r_{t+1}^k - \delta$$

Borrows a_t from HH at rate r_t $\longrightarrow$ "Turns" it into K $\longrightarrow$ rents to firms at r_t^k

r_t^k generates depreciation

$$\Pi^F = (1+r_t^k)K_t - (1+r_t)a_t - \delta \cdot K_t$$

S.t: $K_t = a_t \rightsquigarrow$ Prod. function

Under Perfect competition:

$$\Pi^F = 0 \longrightarrow \boxed{(1+r_t^k - \delta) = 1+r_t}$$

(d)

$$C_t + I_t = A K_t^\alpha$$

$$K_{t+1} = (1-\delta)K_t + I_t$$

$$\left. \begin{array}{l} C_t + I_t = A K_t^\alpha \\ K_{t+1} = (1-\delta)K_t + I_t \end{array} \right\} C_t + K_{t+1} = A K_t^\alpha + (1-\delta)K_t$$

Euler

$$\frac{C_{t+1}}{C_t} = [\beta(1+r)]^{\frac{1}{\sigma}}$$